

Error Analysis for Solving a Linear System by Cholesky Factorization

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Abstract

Following Demmel (1981) with some lemmas improved by Rump and Jeannerod (2014) and previously implemented in libValidSDP by Roux (2017), we derive the backward error analysis for solving a s.d.p. linear system $Ax = b$ with fully accurate consideration of underflows. We do this as a guide to what will be mechanized in Rocq, or what will be documentation of what will have been mechanized.

1 Preliminaries

Almost all of this section is quoted directly from Demmel (1981).

Let ϵ be the rounding error of the arithmetic, and λ be the underflow threshold. Suppose, for example, a binary floating point number is represented as $f \cdot 2^e$ where f lies between 1 and 2 with an n bit fraction, and e is an integer exponent in the range $e_{\min} \leq e \leq e_{\max}$. Then $\epsilon = 2^{-n}$ (the difference between 1 and the next largest number). The underflow threshold is $\lambda = 2^{e_{\min}}$; this is the smallest ... normalized number for G.U. ... Denormalized numbers lie between 0 and $2^{e_{\min}}$ in G.U. arithmetic for which the smallest nonzero number is thus $\mu = \epsilon\lambda$. We will give a more precise model of rounding and underflow errors in the next section.

Let $\bullet$ be one of the operations $(+, -, *, /)$ and let $\text{fl}(a \bullet b)$ denote the floating point result of the indicated computation. ... To take underflow into account, we write ([Kahan and Palmer, 1979])

$$\text{fl}(a \bullet b) = (a \bullet b)(1 + e) + \eta \quad \text{unless } a \bullet b \text{ overflows.} \quad (11)$$

In the case of G.U. we have the following constraints on e and η :

- (1) $|e| \leq \epsilon$ and $|\eta| \leq \lambda\epsilon$,
- (2) $\eta \times e = 0$
- (3) $\eta = 0$ if $\bullet$ is either addition or subtraction.

We define the function $UN(x)$ to be

$$UN(x) = \begin{cases} 1 & \text{if } x \text{ underflows } (x < \lambda) \\ 0 & \text{if it does not} \end{cases}$$

$\|A\|$ and $\|b\|$ denote the norms of matrix A and vector b . $\|A\|_{\infty}$ denotes the infinity norm of A , and similarly for $\|b\|_{\infty}$. $k(A) = \|A\|_{\infty}\|A^{-1}\|_{\infty}$ denotes the condition number of the matrix A .

$|A|$ ($|b|$) denotes the matrix (vector) whose entries are the absolute values of the corresponding entries in A (b). Inequalities like $|A| > |B|$ ($|a| > |b|$) are meant componentwise.

Approach. We use backward error analysis. Thus, when trying to solve

$$Ax = b \quad (12)$$

for x we obtain an approximation $\hat{x} = x + \delta x$ which satisfies the perturbed problem

$$(A + \delta A)\hat{x} = b + \delta b. \quad (13)$$

The task of backwards error analysis is to obtain bounds on δA and δb . These bounds can be used in turn to bound the residual r :

$$r = A\hat{x} - b = -\delta A\hat{x} + \delta b \quad (14)$$

2 Comparison with Demmel (1981)

Demmel's Proposition 2 is as follows. As we explain in the next sections, we'll use something slightly different.

To analyze solving a triangular system, we need another expression for the error in computing an inner product:

$$\text{float}(\sum_{i=1}^n a_i b_i) = \sum_{i=1}^n a_i b_i (1 + E_i) + \eta . \quad (31)$$

In the absence of underflow we have

$$\begin{aligned} |E_1| &\leq n\epsilon \\ |E_j| &\leq (n+2-j)\epsilon \quad \text{if } j > 1 . \end{aligned} \quad (32)$$

In the case of G.U.¹ we have the same bounds on the $|E_i|$, and

$$|\eta| \leq n\lambda\epsilon \quad (34)$$

Demmel's Lemma 4: Error Analysis of Solving a Triangular System. As we explain in the next sections, we make some adjustments to the statement of this lemma.

Consider the lower triangular system $Ly = b$ where L does not necessarily have ones on the diagonal. (The analyses when only ones are on the diagonal or L is upper triangular are similar.) Then in solving $Ly = b$ using the program in Appendix 2 (where inner products are accumulated with roundoff error ϵ_s and smallest number μ_s we obtain $\hat{y} = y + \delta y$ which actually satisfies $(L + \delta L)\hat{y} = b + \delta b$, where

$$|\delta L_{ij}| \leq |L_{ij}| \cdot \begin{cases} \epsilon & \text{if } i = j = 1 \\ (i-1)\epsilon_s & \text{if } i > j = 1 \\ (i+1-j)\epsilon_s & \text{if } i > j > 1 \\ \epsilon + \epsilon_s & \text{if } i = j > 1 \end{cases} \quad (\text{demmel.80})$$

independent of underflow.

In the absence of underflow

$$\delta b = 0 . \quad (\text{demmel.81})$$

Using G.U.

$$\begin{aligned} \text{if } i = 1 \quad |\delta b_i| &\leq \lambda\epsilon |L_{11}| UN(y_1) && \text{if } y_1 \text{ underflows} \\ \text{if } i > 1 \quad |\delta b_i| &\leq i \cdot \lambda_s \epsilon_s + \lambda\epsilon |L_{ii}| UN(y_i) && \begin{array}{l} \text{intermediate underflows} \\ \text{if } y_i \text{ underflows} \end{array} \end{aligned} \quad (\text{demmel.82})$$

If L_{ii} is known to be 1, the only changes in the above bounds (besides substituting 1 for L_{ii}) are that $E_{11} = \delta b_1 = 0$ independent of underflow.

3 Error analysis of Cholesky Solve

Instead of Demmel's Proposition 2, we will be using the following lemma:

Rump/Jeannerod Lemma 2.1: Error analysis of $(c - \sum a_i b_i)/b_k$. (libvalidsdp.cholesky.lemma_2_1)
Let $\hat{y} = fl((c - \sum_{0 \leq i < k} a_i b_i)/b_k)$ computed by repeated subtraction left to right. Assume $b_k \neq 0$. Then

$$|b_k \hat{y} - (c - \sum_{0 \leq i < k} a_i b_i)| < (k+1)\epsilon |b_k \hat{y}| \sum_{0 \leq i < k} |a_i b_i| + (1 + (k+1)\epsilon)(k+1 + |b_k|)\lambda\epsilon. \quad (\text{lemma2.1})$$

Demmel's Lemma 4 provides fine-tuned bounds for four different cases, depending on whether $i > j$ and whether $j > 0$. In all these cases we will prove a weaker bound because that allows us to re-use the Rump/Jeannerod Lemma 2.1 for $c - \sum x_i y_i$.

¹Editor's note: G.U. = Gradual Underflow, that is, a floating-point format that supports subnormal numbers. The technical report also covers the case of "S.Z." or "store zero", which we omit here in all lemmas and theorems.

Theorem Demmel_L4_adapted:

$$\begin{aligned}
|\delta L_{ij}| &\leq |L_{ij}|n\epsilon \quad \text{and} \quad (\text{demmel_lemma_4}) \\
\text{if } i = 0 \quad |\delta b_i| &\leq \lambda\epsilon|L_{00}|UN(y_0) \quad \text{if } y_0 \text{ underflows} \\
\text{if } i > 0 \quad |\delta b_i| &\leq (1 + (i + 1)\epsilon)(i + 1) \cdot \lambda_s\epsilon_s + \text{intermediate underflows} \quad . \quad (1) \\
&\quad (1 + (i + 1)\epsilon)\lambda\epsilon|L_{ii}|UN(y_0) \quad \text{if } y_i \text{ underflows}
\end{aligned}$$

The factor $(1 + (i + 1)\epsilon)$ was omitted from Demmel's original but is required for correctness.

Proof. From libValidSDP lemma_2_1_aux, by a derivation of several lines. Q.E.D.

Solving a linear system by Cholesky+Substitution

Let $\hat{x}$ be the result of solving $Ax = b$ by Cholesky decomposition $A = R^T R$ followed by forward substitution $R^T y = b$ then back substitution $Rx = y$.

Lemma th_2_3_aux3: Accuracy of Cholesky decomposition. Suppose A is an $n \times n$ symmetric positive definite floating-point matrix, with $(n + 1)\epsilon < 1$. Let $\hat{R}$ be the result of floating-point Cholesky decomposition (inner-product method) of A into $\hat{R}^T \hat{R}$. Let $\delta A = A - \hat{R}^T \hat{R}$. Let the vector R_j be the j th column of R . Let $a_{\max}$ be the maximum diagonal element of $|A|$. Then

$$|\delta A_{ij}| < (\min(i, j) + 2)\epsilon \|R_i\|_2 \|R_j\|_2 + 4\lambda\epsilon(n + 1 + a_{\max}). \quad (2.3aux3)$$

Proof. See Rump+Jeannerod, or see libValidSDP. Q.E.D.

Remark. Let A and $\hat{R}$ be as above. Then

$$\|\hat{R}_j\|_2 \leq \sqrt{(A_{jj} + 2j\lambda\epsilon)/(1 - (j + 2)\epsilon)}. \quad (2.3aux1)$$

Proof. LibValidSDP th_2_3_aux1.

Corollary. Combining (2.3aux1) and (2.3aux2), we have,

$$\begin{aligned}
|\delta A_{ij}| &< (\min(i, j) + 2)\epsilon \sqrt{(A_{ii} + 2i\lambda\epsilon)/(1 - (i + 2)\epsilon)} \sqrt{(A_{jj} + 2j\lambda\epsilon)/(1 - (j + 2)\epsilon)} + 4\lambda\epsilon(n + 1 + a_{\max}) \\
&\leq (n + 1)\epsilon \sqrt{(a_{\max} + 2(n - 1)\lambda\epsilon)/(1 - (n + 1)\epsilon)} \\
&\quad \cdot \sqrt{(a_{\max} + 2(n - 1)\lambda\epsilon)/(1 - (n + 1)\epsilon)} + 4\lambda\epsilon(n + 1 + a_{\max}) \\
&= (n + 1)\epsilon(a_{\max} + 2(n - 1)\lambda\epsilon)/(1 - (n + 1)\epsilon) + 4\lambda\epsilon(n + 1 + a_{\max}) \\
&\quad (\text{ch_back_err})
\end{aligned}$$

Lemma. We obtain bounds on $\delta R^T \hat{R}$ (et cetera) by combining (2.3aux1) and (demmel_lemma_4):

$$\begin{aligned}
|(\delta R^T \hat{R})_{ij}| &\leq (|\delta R|^T |\hat{R}|)_{ij} \\
&\leq \sum_k (k + 1)\epsilon |\hat{R}_{ki}| \cdot |\hat{R}_{kj}| \quad (\text{by Lemma 0.4}) \\
&\leq n\epsilon \|\hat{R}_i\|_2 \|\hat{R}_j\|_2 \quad (\text{by Cauchy - Schwartz}) \\
&\leq n\epsilon \sqrt{(A_{ii} + 2i\lambda\epsilon)/(1 - (i + 2)\epsilon)} \sqrt{(A_{jj} + 2j\lambda\epsilon)/(1 - (j + 2)\epsilon)} \quad (2.3aux1) \\
&\leq n\epsilon \sqrt{(a_{\max} + 2(n - 1)\lambda\epsilon)/(1 - (n + 1)\epsilon)} \sqrt{(a_{\max} + 2(n - 1)\lambda\epsilon)/(1 - (n + 1)\epsilon)} \\
&= n\epsilon(a_{\max} + 2(n - 1)\lambda\epsilon)/(1 - (n + 1)\epsilon) \\
&\quad (\text{csolve_back_err_aux1})
\end{aligned}$$

Lemma. Bounds on $\delta R^T \delta R$ as follows:

$$\begin{aligned}
|(\delta R^T \delta R)_{ij}| &\leq (|\delta R|^T |\delta R|)_{ij} \\
&\leq \sum_k |\delta R^T|_{ik} |\delta R|_{kj} \\
&= \sum_k |\delta R|_{ki} |\delta R|_{kj} \\
&\leq \sum_k (k + 1)^2 \epsilon^2 |\hat{R}_{ki}| \cdot |\hat{R}_{kj}| \quad (\text{csolve_back_err_aux2}) \\
&\leq n^2 \epsilon^2 \|\hat{R}_i\|_2 \|\hat{R}_j\|_2 \\
&\leq n^2 \epsilon^2 (a_{\max} + 2(n - 1)\lambda\epsilon)/(1 - (n + 1)\epsilon).
\end{aligned}$$

Theorem: Cholesky factor + substitution. Suppose A is an $n \times n$ symmetric positive definite floating-point matrix with minimum eigenvalue $> \sqrt{\lambda}$, with $(n+1)\epsilon < 1$. Suppose b is a floating-point vector of length n . Suppose $\hat{x}$ is the result of floating-point Cholesky factorization of A into $\hat{R}^T \hat{R}$ followed by forward substitution finding $\hat{y}$ such that $\hat{R}^T \hat{y} = b$ and then $\hat{R}\hat{x} = \hat{y}$. Taking into account underflows in the factorization, but not yet considering underflows in the substitutions, we have that there is a backward error δA such that $(A + \delta A)\hat{x} = b$, bounded as follows:

$$|\delta A| \leq ((3n+1)\epsilon + n^2\epsilon^2) \frac{a_{\max} + 2(n-1)\lambda\epsilon}{1 - (n+1)\epsilon} + 4\lambda\epsilon(n+1+a_{\max}). \quad (\text{csolve_back_err})$$

Proof. If it were not for the underflow terms δb and δy , we would have

$$\begin{aligned} b &= (\hat{R}^T + \delta R^T)\hat{y} \\ &= (\hat{R}^T + \delta R^T)(\hat{R} + \delta R)\hat{x} \\ &= (\hat{R}^T \hat{R} + \delta R^T \hat{R} + \hat{R}^T \delta R + \delta R^T \delta R)\hat{x} \\ &= (A + \delta A' + \delta R^T \hat{R} + \hat{R}^T \delta R + \delta R^T \delta R)\hat{x} \\ &= A\hat{x} + (\delta A' + \delta R^T \hat{R} + \hat{R}^T \delta R + \delta R^T \delta R)\hat{x}. \end{aligned} \quad (\text{xxx})$$

We will define

$$\begin{aligned} \delta A &= \delta A' + \delta R^T \hat{R} + \hat{R}^T \delta R + \delta R^T \delta R \\ &\leq |\delta A'| + |\delta R^T \hat{R}| + |\hat{R}^T \delta R| + |\delta R^T \delta R|. \end{aligned} \quad (\text{xxx})$$

From (2.3aux3) we have $|\delta A'_{ij}| < (\min(i, j) + 2)\epsilon \|R_i\|_2 \|R_j\|_2 + 4\lambda\epsilon(n+1+a_{\max})$.

From csolve_back_err_aux1 we have $|(\delta R^T \hat{R})_{ij}| \leq n\epsilon(a_{\max} + 2(n-1)\lambda\epsilon)/(1 - (n+1)\epsilon)$

and similarly $|(\hat{R}^T \delta R)_{ij}| \leq n\epsilon(a_{\max} + 2(n-1)\lambda\epsilon)/(1 - (n+1)\epsilon)$.

From csolve_back_err_aux2 we have $|\delta R^T \delta R| \leq n^2\epsilon^2(a_{\max} + 2(n-1)\lambda\epsilon)/(1 - (n+1)\epsilon)$.

The result follows immediately. Q.E.D.

Still to do: Account for underflows in the forward substitution and back substitution.